Complex Manifolds

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Abstract

Contents

1 Complex Manifolds

1.1 Basic Definitions

Definition 1.1.1 (The Holomorphic Pseudogroup of Transformations)
Define the holomorphic Pseudogroup of transformations to be

$$\Gamma_{\text{Holo}} = \{ f : U \to V \mid f \text{ is biholomorphic and } U, V \subseteq \mathbb{C}^n \text{ are open } \}$$

Definition 1.1.2 (Complex Manifolds)
A complex manifold M is a manifold whose atlas is given by Γ_{Holo}.

Proposition 1.1.3 Every complex manifold is a smooth real manifold.

Proof Let $(U, \phi = z_1, \dots, z_n))$ be a chart of a complex manifold M. Write $z_k = x_k + iy_k$ where x_k and y_k are smooth real valued functions on U. Then $(x_1, \dots, x_n, y_1, \dots, y_n)$ gives a homeomorphism between U and an open subset of $\mathbb{R}^{2n}$. ∎

Lemma 1.1.4
Let M be a complex manifold. Let $U \subseteq M$ be an open subset. Then U is a complex manifold.

Definition 1.1.5 (Almost Complex Manifolds)
An almost complex manifold is a smooth manifold M together with an almost complex structure on TM.

Even though every even dimensional vector space has a linear complex structure, they may not glue smoothly into an almost complex structure for a smooth manifold. This is a question of the reduction of the structure group from $GL_{2n}(\mathbb{R})$ to the subgroup $GL_n(\mathbb{C})$.

1.2 The Sheaf of Holomorphic Forms

Definition 1.2.1 (The Sheaf of Holomorphic Forms)
Let M be a complex manifold. Define the sheaf of holomorphic form $\mathcal{O}_M$ as follows.
- For each open set $U \subseteq M$, define $\mathcal{O}_M(U) = \{ f : U \to \mathbb{C} \mid f \text{ is holomorphic } \}$.
- For each inclusion of open sets $V \hookrightarrow U$, the map $\mathcal{O}_M(U) \to \mathcal{O}_M(V)$ is given by the restriction map $f \mapsto f|_V$.

Lemma 1.2.2
Let M be a compact connected complex manifold. Then we have

$$\mathcal{O}_M(M) \cong \mathbb{C}$$

Theorem 1.2.3 (Oka's (Coherence) Theorem)
Let M be a complex manifold. Then $\mathcal{O}_M$ is a coherent sheaf.

Definition 1.2.4 (The Sheaf of Meromorphic Forms)
Let M be a complex manifold. Define the sheaf of meromorphic form $\mathcal{M}_M$ as follows.
- For each open set $U \subseteq M$, define $\mathcal{M}_M(U) = \text{Frac}(\mathcal{O}_M(U))$.
- For each inclusion of open sets $V \hookrightarrow U$, the map $\mathcal{M}_M(U) \to \mathcal{M}_M(V)$ is given by the restriction map $f \mapsto f|_V$.

1.3 Complex Submanifolds

Definition 1.3.1 (Complex Submanifolds) Let M be an n dimensional manifold. An m dimensional submanifold is a subset Y of M such that for every $y \in Y$, there exists a chart (U, ϕ) of M such that

$$\phi(Y \cap U) = \phi(U) \cap \{0\}^{n-m} \times \mathbb{C}^m = \{(z_1, \dots, z_n) \in \phi(U) | z_{n-m+1} = \dots = z_n = 0\}$$

2 More on Vector Bundles

2.1 Holomorphic Vector Bundles

Definition 2.1.1 (Holomorphic Vector Bundles)
Let M be a complex manifold. A holomorphic vector bundle is a vector bundle $p : E \to B$ over $\mathbb{C}$ such that the following are true.

- E is a complex manifold.
- p is holomorphic.
- The local trivialization charts $\phi : p^{-1}(U) \xrightarrow{\cong} U \times \mathbb{C}^r$ are biholomorphic.

Proposition 2.1.2
Let M be a complex manifold. Let E be a complex manifold and $p : E \to M$ a vector bundle over $\mathbb{C}$. Then p is a holomorphic vector bundle if and only if the transition maps $g_{i,j} : U_i \cap U_j \to \text{GL}(n, \mathbb{C})$ are holomorphic.

2.2 Vector Bundles and Sheaves

Theorem 2.2.1 (Serre's Correspondence for Topological Manifolds)
Let M be a topological manifold. Then there is an equivalence of categories

$$\textbf{Vect}^{\mathbb{R}}(M) \simeq \textbf{LF}_{\mathcal{C}^0_M}$$

Theorem 2.2.2 (Serre's Correspondence for Smooth Manifolds)
Let M be a smooth manifold. Then there are equivalence of categories

$$\textbf{SmVect}^{\mathbb{R}}(M) \simeq \textbf{LF}_{\mathcal{C}^\infty_M} \quad \text{and} \quad \textbf{SmVect}^{\mathbb{C}}(M) \simeq \textbf{LF}_{\mathcal{C}^\infty_M \otimes_{\mathbb{R}} \mathbb{C}}$$

Theorem 2.2.3 (Serre's Correspondence for Complex Manifolds)
Let M be a complex manifold. Then there is an equivalence of categories

$$\textbf{HolVect}^{\mathbb{C}}(M) \simeq \textbf{LF}_{\mathcal{O}_M}$$

Theorem 2.2.4 (Serre-Swan Theorem for Manifolds)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle over M. Then the following are true.

- $\mathcal{C}^\infty(E)$ is a finitely generated and projective $\mathcal{C}^\infty(M)$-module.
- Suppose that M is connected. There is an equivalence of categories

$$\mathcal{C}^\infty : \textbf{SmVect}^{\mathbb{R}}(M) \xrightarrow{\simeq} \textbf{proj}(\mathcal{C}^\infty(M))$$

Theorem 2.2.5 (Serre-Swan Theorem)
Let X be a compact Hausdorff space. Then there are equivalence of categories

$$\Gamma : \textbf{Vect}^{\mathbb{R}}(X) \xrightarrow{\simeq} \textbf{proj}(\mathcal{C}^0(X)) \quad \text{and} \quad \Gamma : \textbf{Vect}^{\mathbb{C}}(X) \xrightarrow{\simeq} \textbf{proj}(\mathcal{C}^0(X) \otimes_{\mathbb{R}} \mathbb{C})$$

3 Tangent Spaces

3.1 The Complex Structure of Tangent Spaces on Almost Complex Manifolds

Definition 3.1.1 (Holomorphic and Antiholomorphic Tangent Space)
Let (M,J) be an almost complex manifold.
- Define $T_p^{1,0}(M)$ to be the eigenspace of i of the linear complex structure J_p on $T_{\mathbb{C},p}(M)$.
- Define $T_p^{0,1}(M)$ to be the eigenspace of $-i$ of the linear complex structure J_p on $T_{\mathbb{C},p}(M)$.

We have seen above that
$$T_pM\otimes_R\mathbb{C}\cong T_p^{(1,0)}(M)\oplus T_p^{(0,1)}(M)$$
as complex vector spaces.

Lemma 3.1.2
Let (M,J) be an almost complex manifold. Let $p\in M$. Let $(U,x^1,\dots,x^n,y^1,\dots,y^n)$ be a smooth chart at p. Suppose that $J_p\left(\left.\frac{\partial}{\partial x^i}\right|_p\right)=\left.\frac{\partial}{\partial y^i}\right|_p$ for $1\leq i\leq n$. Then the following are true.
- A $\mathbb{C}$-basis for $T_p^{1,0}(M)$ is given by
$$\left\{\frac{1}{2}\left.\left(\frac{\partial}{\partial x^1}-i\frac{\partial}{\partial y^1}\right)\right|_p,\dots,\frac{1}{2}\left.\left(\frac{\partial}{\partial x^n}-i\frac{\partial}{\partial y^n}\right)\right|_p\right\}$$
- A $\mathbb{C}$-basis for $T_p^{0,1}(M)$ is given by
$$\left\{\frac{1}{2}\left.\left(\frac{\partial}{\partial x^1}+i\frac{\partial}{\partial y^1}\right)\right|_p,\dots,\frac{1}{2}\left.\left(\frac{\partial}{\partial x^n}+i\frac{\partial}{\partial y^n}\right)\right|_p\right\}$$

Assuming that J_p sends the basis vector $\left.\frac{\partial}{\partial x^i}\right|_p$ to $\left.\frac{\partial}{\partial y^i}\right|_p$ loses no generality since we can always choose appropriate coordinate functions so that the almost complex structure locally at U looks like ours one.

Recall from complex analysis that we defined partial complex derivatives by
$$\frac{\partial}{\partial z_k}=\frac{1}{2}\left(\frac{\partial}{\partial x_k}-i\frac{\partial}{\partial y_k}\right)\quad\text{and}\quad\frac{\partial}{\partial\overline{z_k}}=\frac{1}{2}\left(\frac{\partial}{\partial x_k}+i\frac{\partial}{\partial y_k}\right)$$

Definition 3.1.3 (The Basis for Tangent Space given by the Almost Complex Structure)
Let (M,J) be an almost complex manifold. Let $p\in M$. Let $(U,x^1,\dots,x^n,y^1,\dots,y^n)$ be a smooth chart at p. Define the tangent vector
$$\left.\frac{\partial}{\partial z^k}\right|_p:\mathcal{C}^\infty_{M,p}\to\mathbb{R}$$
by
$$\left.\frac{\partial}{\partial z^k}\right|_p=\left.\left(\frac{\partial}{\partial x^k}-i\frac{\partial}{\partial y^k}\right)\right|_p$$
Define the tangent vector
$$\left.\frac{\partial}{\partial\overline{z^k}}\right|_p:\mathcal{C}^\infty_{M,p}\to\mathbb{R}$$
by
$$\left.\frac{\partial}{\partial\overline{z^k}}\right|_p=\left.\left(\frac{\partial}{\partial x^k}+i\frac{\partial}{\partial y^k}\right)\right|_p$$

3.2 Decomposing the Tangent Space of Complex Manifolds

Let M be a complex manifold. Let $p \in M$. Regarded as a smooth manifold, we have the notion of tangent space $T_pM = \text{Der}(C^\infty_{M,p}(M,\mathbb{R}),\mathbb{R})$. Since we are on a complex manifold, we ought to consider smooth functions with values in $\mathbb{C}$ instead.

Lemma 3.2.1
Let M be a complex manifold. Let $p \in M$. Let $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ be a smooth chart at p. Then an $\mathbb{R}$-basis for T_pM is given by

$$\left\{ \frac{\partial}{\partial x^1}\bigg|_p, \dots, \frac{\partial}{\partial x^n}\bigg|_p, \frac{\partial}{\partial y^1}\bigg|_p, \dots, \frac{\partial}{\partial y^n}\bigg|_p \right\}$$

Definition 3.2.2 (The Complex Tangent Space)
Let M be a complex manifold. Let $p \in M$. Define the complex tangent space of M at p to be

$$T_{\mathbb{C},p}(M) = \text{Der}_{\mathbb{C}}(C^\infty_{M,p,\mathbb{C}},\mathbb{C})$$

Let $u : \mathbb{C} \to \mathbb{C}$ be a function. Identifying $\mathbb{C}$ with $\mathbb{R}^2$, we obtain the notion of smooth functions. And smoothness of u can be checked on the components of u. This gives a decomposition of tangent spaces.

Lemma 3.2.3
Let M be a complex manifold. Let $p \in M$. Then we have a natural $\mathbb{C}$-vector space isomorphism

$$T_{\mathbb{C},p}(M) \cong T_p(M) \otimes_{\mathbb{R}} \mathbb{C}$$

In particular, the real dimension of $T_{\mathbb{C},p}(M)$ is $4\dim_{\mathbb{C}}(M)$, and a $\mathbb{C}$-basis for $T_{\mathbb{C},p}(M)$ is given by lemma 3.5.1. This is the effect of considering $T_{\mathbb{C},p}(M)$ as an extension of scalars from $T_p(M)$.

Proposition 3.2.4
Let M be a complex manifold. Then M is an almost complex manifold.

We know from Complex Analysis that not every smooth function is holomorphic. A smooth function must satisfy the Cauchy-Riemann equations in order to be holomorphic. We will see in a moment that the complex tangent space contains the holomorphic tangent space defined earlier.

3.3 The Holomorphic Tangent Space

Let M be a smooth manifold. Recall that the tangent space of M is defined to be

$$\text{Der}_{\mathbb{R}}(C^\infty_{M,p}, C^\infty_{M,p}/m_p)$$

where $C^\infty_{M,p}$ is the stalk of the sheaf $C^\infty_M(-)$ of smooth functions on M at p. Now if M is a complex manifold, the canonical maps we care about are holomorphic maps and the sheaf of holomorphic functions $\mathcal{O}_M(-)$. Therefore we define the holomorphic tangent space in a similar way.

Definition 3.3.1 (The Holomorphic Tangent Space)
Let M be a complex manifold. Let $p \in M$. Define the Holomorphic Tangent space of M at p to be

$$T_p^{\text{Holo}}(M) = \text{Der}_{\mathbb{C}}(\mathcal{O}_{M,p},\mathbb{C})$$

where $\mathbb{C}$ is a $\mathcal{O}_{M,p}$-module via the isomorphism $\mathbb{C} \cong \frac{\mathcal{O}_{M,p}}{m_p}$.

Immediately one is curious of the relation between $T_p^{\text{Holo}}(M)$ and $T_p(M)$ since every complex manifold is a smooth manifold of double the dimension.

Proposition 3.3.2
Let M be a complex manifold. Let $p \in M$. Then we have $T_p^{1,0}(M) = T_p^{\text{Holo}}(M)$.

The way in which we may decompose the complexified tangent space into its holomorphic and anti-holomorphic part works also for almost complex manifolds. In fact this motivates the theory for almost complex manifolds in the first place.

3.4 The Induced Map and the Jacobians

Let M, N be complex manifolds. Let $f : M \to N$ be a smooth map. Let $p \in M$. Let $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ be a smooth chart at p. Let $(V, u^1, \dots, u^m, v^1, \dots, v^m)$ be a smooth chart at $f(p)$. Recall that the $\mathbb{R}$-linear map $(f_*)_p : T_p(M) \to T_p(N)$ is given by the Jacobian which can be compactly written into block matrix form

$$(f_*)_p = \begin{pmatrix} \frac{\partial u^i \circ f}{\partial x^j}\Big|_p & \frac{\partial u^i \circ f}{\partial y^j}\Big|_p \\ \frac{\partial v^i \circ f}{\partial x^j}\Big|_p & \frac{\partial v^i \circ f}{\partial y^j}\Big|_p \end{pmatrix}$$

We may ask what the Jacobian looks like under our newly acquired basis from the almost complex structure.

Definition 3.4.1 (The Holomorphic Jacobian)
Let M, N be complex manifolds. Let $f : M \to N$ be a smooth map. Let $p \in M$. Let $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ be a smooth chart at p. Let $(V, u^1, \dots, u^m, v^1, \dots, v^m)$ be a smooth chart at $f(p)$. Define the holomorphic Jacobian of f with respect to the charts to be

$$J(f) = \begin{pmatrix} \frac{\partial (u^1 + iv^1) \circ f}{\partial z^1}\Big|_p & \cdots & \frac{\partial (u^1 + iv^1) \circ f}{\partial z^n}\Big|_p \\ \vdots & \ddots & \vdots \\ \frac{\partial (u^m + iv^m) \circ f}{\partial z^1}\Big|_p & \cdots & \frac{\partial (u^m + iv^m) \circ f}{\partial z^n}\Big|_p \end{pmatrix}$$

Proposition 3.4.2
Let M, N be complex manifolds. Let $f : M \to N$ be a smooth map. Let $p \in M$. Let $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ be a smooth chart at p. Let $(V, u^1, \dots, u^m, v^1, \dots, v^m)$ be a smooth chart at $f(p)$. Then the differential $(f_*)_p$ under the basis induced from the almost complex structure is given in block matrix form by

$$(f_*)_p = \begin{pmatrix} J(f) & 0 \\ 0 & \overline{J(f)} \end{pmatrix}$$

Proposition 3.4.3
Let M, N be complex manifolds. Let $f : M \to N$ be a holomorphic map. Let $p \in M$. Let $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ be a smooth chart at p. Let $(V, u^1, \dots, u^m, v^1, \dots, v^m)$ be a smooth chart at $f(p)$. Then $(f_*)_p : T_p(M) \to T_p(N)$ restricts to an $\mathbb{C}$-linear map $T_p^{1,0}(M) \to T_{f(p)}^{1,0}(N)$.

3.5 The Complex Tangent Bundle

Definition 3.5.1 (The Complex Tangent Bundle)
Let M be a complex manifold. Define the complex tangent bundle of M to be

$$T_{\mathbb{C}} M = TM \otimes_{\mathbb{R}} \mathbb{C}$$

3.6 Integrable Complex Structures

Definition 3.6.1 (Integrable Complex Structure)
Let (M, J) be an almost complex manifold. We say that M is integrable if there exist a complex structure on M that induces J.

Theorem 3.6.2 (The Newlander–Nirenberg theorem)
Let (M, J) be an almost complex manifold. Then J is integrable if and only if $[TM^-, TM^-] \subseteq TM^-$

4 Complex Differential Forms

4.1 Complexified Differential Forms

Definition 4.1.1
Let M be a complex manifold. Define ??? to be

$$\Omega^k_{M,\mathbb{C}}$$

to be the sheaf of smooth sections of $\Lambda^k(T_\mathbb{C}M)$.

Let M be a complex manifold. The $\mathbb{C}$-vector space $\Omega^1_\mathbb{C}(M)$ consists of smooth maps from M to $T^*_\mathbb{C}M$. The splitting $T^*_\mathbb{C}M = TM^+ \oplus TM^-$ together with sections on operations of vector bundles means that giving a section of $T^*_\mathbb{C}M$ is the same as giving a section of TM^+ and TM^-. Thus we acquire a splitting $\Omega^1_\mathbb{C}(M) = \Omega^{1,0}(M) \oplus \Omega^{0,1}(M)$.

Let M be a complex manifold. The splitting $T^*_\mathbb{C}M = TM^+ \oplus TM^-$. Properties of the exterior product that translates to vector bundles as

$$\Lambda^n(T^*_\mathbb{C}M) = \bigoplus_{p+q=n} \Lambda^p(TM^+) \otimes \Lambda^q(TM^-)$$

Definition 4.1.2
Let M be a complex manifold. Define ??? to be

$$\Omega^{p,q}_M$$

to be the sheaf of smooth sections of $\Lambda^p(TM^+) \otimes \Lambda^q(TM^-)$.

It follows that $\Omega^n_\mathbb{C}(M) = \bigoplus_{p+q=n} \Omega^{p,q}(M)$.

Proposition 4.1.3
Let M be a complex manifold. Then the following are true.

- There is a $\mathcal{C}^\infty_M$-module isomorphism

$$\Omega^n_{M,\mathbb{C}} \cong \bigoplus_{p+q=n} \Omega^{p,q}_M$$

- There is a $\mathcal{C}^\infty_M$-module isomorphism

$$\Omega^{p,q}_M \cong \Lambda^p\Omega^{1,0}_M \otimes_\mathbb{C} \Lambda^q\Omega^{0,1}_M$$

Definition 4.1.4 (The Complexified Differential)
Let M be a complex manifold.

- Define the differential $d_\mathbb{C} : \Omega^\bullet_{M,\mathbb{C}}(M) \to \Omega^\bullet_{M,\mathbb{C}}(M)$ of M to be the $\mathcal{C}^\infty_{M,\mathbb{C}}(M)$-module homomorphism

$$d_\mathbb{C} = d \otimes \text{id}_\mathbb{C}$$

  where $d : \Omega^\bullet(M) \to \Omega^\bullet(M)$ is the differential of M considered as a smooth manifold.
- Define the $\mathcal{C}^\infty_{M,\mathbb{C}}$-module homomorphism $d_\mathbb{C} : \Omega^\bullet_{M,\mathbb{C}} \to \Omega^\bullet_{M,\mathbb{C}}$ as follows. For each open set U, the $\mathcal{C}^\infty_{M,\mathbb{C}}(U)$-module homomorphism $d_\mathbb{C}(U)$ is the differential of U considered as a complex manifold.

Definition 4.1.5 (The Complex de Rham Chain Complex)
Let M be a complex manifold. Define the de Rham chain complex of M to be $(\Omega^\bullet_{M,\mathbb{C}}(M), d_\mathbb{C})$.

Definition 4.1.6 (The Complex de Rham Cohomology Groups)
Let M be a complex manifold. Define the nth holomorphic de Rham cohomology groups of M to

be
$$H^n_{\mathrm{dR}}(M;\mathbb{C}) = H^n(\Omega^\bullet_{M,\mathbb{C}}(M), d_{\mathbb{C}})$$

Proposition 4.1.7
Let M be a complex manifold. Then there is a natural isomorphism
$$H^n_{\mathrm{dR}}(M;\mathbb{C}) \cong H^n_{\mathrm{dR}}(M;\mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$$
for all $n \in \mathbb{N}$.

4.2 Dolbeault Cohomology

Definition 4.2.1 (Dolbeault Operators)
Let M be a complex manifold. Define the Dolbeault operators on M by
$$\partial = \mathrm{proj}_{p+1,q} \circ d : \Omega^{p,q} \to \Omega^{p+1,q} \quad \text{and} \quad \overline{\partial} = \mathrm{proj}_{p,q+1} \circ d : \Omega^{p,q} \to \Omega^{p,q+1}$$

Proposition 4.2.2
Let M be a complex manifold. Let $p \in M$. Let $(U, z^1, \dots, z^n)$ be a local chart at p. Suppose that $\omega \in \Omega^{p,q}_M$ is given locally by
$$\omega = \sum_{|I|=p,|J|=q} f_{I,J} dz^I \wedge d\overline{z^J}$$
where $f_{I,J} : U \to \mathbb{C}$ are smooth functions. Then the following are true.
- $\partial\omega$ is given locally on the same chart by
$$\partial\omega = \sum_{|I|=p,|J|=q} \sum_k \frac{\partial f_{I,J}}{\partial z^l} dz^l \wedge dz^I \wedge \overline{z^J}$$
- $\overline{\partial}\omega$ is given locally on the same chart by
$$\overline{\partial}\omega = \sum_{|I|=p,|J|=q} \sum_k \frac{\partial f_{I,J}}{\partial \overline{z^l}} d\overline{z^l} \wedge dz^I \wedge \overline{z^J}$$

Proposition 4.2.3
Let M be a complex manifold. Then the following are true.
- $d = \partial + \overline{\partial}$.
- $\partial \circ \partial = \overline{\partial} \circ \overline{\partial} = 0$.
- $\partial \circ \overline{\partial} = -\overline{\partial} \circ \partial$.
- Let $\omega \in \Omega^k_{M,\mathbb{C}}(M)$ and $\eta \in \Omega^l_{M,\mathbb{C}}(M)$. Then we have
$$\partial(\omega \wedge \eta) = \partial(\omega) \wedge \eta + (-1)^k \omega \wedge \partial(\eta) \quad \text{and} \quad \overline{\partial}(\omega \wedge \eta) = \overline{\partial}(\omega) \wedge \eta + (-1)^k \omega \wedge \overline{\partial}(\eta)$$

We can now write the bicomplex in a grid:

$$\begin{tikzcd}
\vdots & \vdots & \vdots & \\
\Omega^{0,2}(M) \arrow[r, "\partial"] \arrow[u] & \Omega^{1,2}(M) \arrow[r, "\partial"] \arrow[u] & \Omega^{2,2}(M) \arrow[r] \arrow[u] & \cdots \\
\Omega^{0,1}(M) \arrow[r, "\partial"] \arrow[u, "\overline{\partial}"] & \Omega^{1,1}(M) \arrow[r, "\partial"] \arrow[u, "\overline{\partial}"] & \Omega^{2,1}(M) \arrow[r] \arrow[u, "\overline{\partial}"'] & \cdots \\
\mathcal{O}_M(M) \arrow[r, "\partial"] \arrow[u, "\overline{\partial}"] & \Omega^{1,0}(M) \arrow[r, "\partial"] \arrow[u, "\overline{\partial}"] & \Omega^{2,0}(M) \arrow[r] \arrow[u, "\overline{\partial}"'] & \cdots
\end{tikzcd}$$

Note that the squares here are not commutative, and this is not the notation we have defined for bicomplexes. If we augment the differential in the total complex by accounting the anti-commutativity of the squares, we would obtain isomorphisms

$$H^k_{\text{dR}}(M;\mathbb{C}) = H^k(\text{Tot}(\Omega^{\bullet,\bullet}(M),\partial,\overline{\partial}))$$

Definition 4.2.4 (Holomorphic de Rham Cohomology)
Let M be a complex manifold. Define the nth holomorphic de Rham cohomology group of M to be

$$H^n_{\text{Holo}}(M) = H^n(\Omega^{\bullet,0}_M(M),\partial)$$

Definition 4.2.5 (Dolbeault Complex) Let M be a complex manifold. Let $p\in\mathbb{N}$. Define the pth Dolbeault complex of M to be the cochain complex

$$\cdots \longrightarrow \Omega^{p,q-1}(M) \xrightarrow{\overline{\partial}} \Omega^{p,q}(M) \xrightarrow{\overline{\partial}} \Omega^{p,q+1}(M) \xrightarrow{\overline{\partial}} \cdots$$

denoted $\Omega^{p,\bullet}(M)$.

Definition 4.2.6 (Dolbeault Cohomology) Define the dobeault cohomology of a complex manifold M to be

$$H^{p,q}(M) = \frac{\ker(\bar{\partial}:\Omega^{p,q}(M)\to\Omega^{p,q+1}(M))}{\text{im}(\bar{\partial}:\Omega^{p,q-1}(M)\to\Omega^{p,q}(M))} = H^q(\Omega^{p,\bullet}(M),\overline{\partial})$$

We may relate Dolbeault cohomology with sheaf cohomology.

Theorem 4.2.7
Let M be a complex manifold. Then we have an isomorphism

$$H^{p,q}(M)\cong H^q(M,\Omega^p_M)$$

Corollary 4.2.8
Let M be a compact complex manifold. Then $H^{0,0}(M)=\mathbb{C}$.

Lemma 4.2.9
Let M be a complex manifold. Then $H^{0,q}(M)=0$ for all $q>\dim(M)$.

4.3 Orientability

Proposition 4.3.1
Let M be a complex manifold. Then M is orientable.

Corollary 4.3.2
Let M,N be complex manifolds. Let $f:M\to N$ be a holomorphic map. Then f preserves orientation.

5 Hermitian Manifolds

5.1 The Hermitian Metric

Definition 5.1.1 (Hermitian Manifolds)
Let M be a complex manifold. We say that M is Hermitian if $T^{1,0}(M)$ admits a Hermitian metric.

Proposition 5.1.2
Let (M, h) be a Hermitian manifold. Let $p \in M$. Let $(U, z^1, \dots, z^n)$ be a complex chart at p. Then h_p is given locally on the chart by

$$h_p = \sum_{i,j=1}^{n} h_{i,j}|_p \, (dz^i)_p \otimes (d\overline{z^j})_p$$

where $h_{i,j}|_p = h_p \left(\frac{\partial}{\partial z^i}\bigg|_p, \frac{\partial}{\partial z^j}\bigg|_p \right)$.

Proposition 5.1.3
Let M be a smooth manifold. Let $p : E \to M$ be a complex vector bundle. Then E admits a Hermitian metric.

5.2 The Associated Form

Definition 5.2.1 (The Associated Form)
Let (M, h) be a Hermitian manifold. Define the associated form of h to be

$$\omega = \frac{i}{2}(h - \overline{h})$$

In local coordinates, ω is expressed as

$$\omega = \frac{i}{2} \sum_{\alpha,\beta=1}^{n} h_{\alpha\overline{\beta}} dz_\alpha \wedge d\overline{z}_\beta$$

if M is a complex manifold of complex dimension n.

Definition 5.2.2 (The Associated Riemannian Metric)
Let (M, h) be a Hermitian manifold. Define the associated Riemannian metric of h to be

$$g = \frac{1}{2}(h + \overline{h}) =$$

In local coordinates, g is expressed as

$$g(u, v) = \frac{1}{2} \sum h_{\alpha\overline{\beta}}(dz_\alpha \otimes d\overline{z}_\beta + d\overline{z}_\beta \otimes dz_\alpha)$$

Lemma 5.2.3
Let (M, h) be a Hermitian manifold. Then we have ω is a $(1, 1)$-form.

Lemma 5.2.4
Let (M, h) be a Hermitian manifold. Then we have

$$h = g - i\omega$$

Proposition 5.2.5
Let (M, h) be a Hermitian manifold whose associated almost complex structure is given by J. Then the following are true.
- $h(Ju, Jv) = h(u, v)$
- $g(Ju, Jv) = g(u, v)$
- $\omega(Ju, Jv) = \omega(u, v)$

Proposition 5.2.6
Let (M, h) be a Hermitian manifold whose associated almost complex structure is given by J. Then the following are true.
- $\omega(u, v) = g(Ju, v)$.
- $g(u, v) = \omega(u, Jv)$.

Notice that the three equations

$$h = g - i\omega \quad \text{and} \quad \omega(u, v) = g(Ju, v) \quad \text{and} \quad g(u, v) = \omega(u, Jv)$$

completely determines the triple (g, h, ω).

Proposition 5.2.7
Let M be a complex manifold whose associated almost complex structure is given by J. Let g be a Riemannian metric on M. Then the following are true.
- $h = g - i\omega$ is a Hermitian form on M where $\omega(u, v) = g(Ju, v)$.
- ω is the associated form of h.

Proposition 5.2.8
Let M be a complex manifold whose associated almost complex structure is given by J. Let ω be a $(1, 1)$-form on M that is non-degenerate, and preserves J and is positive definite in the sense that $\omega(u, Jv) > 0$ for all non-zero u, v. Then the following are true.
- $g(u, v) = \omega(u, Jv)$ is a Riemannian metric on M.
- $h = g - i\omega$ is a Hermitian metric on M.

Proposition 5.2.9
Let (M, h) be a complex Hermitian manifold. Let ω be its associated 2-form. Then the volume form of M is given by

$$dV = \frac{1}{n!}\omega^n$$

6 Complex Hodge Theory

Let (M, h) be a (almost) complex manifold. Then M is necessarily an orientable Riemannian manifold as seen above. Therefore Hodge theory applies to M. In particular, if d is the exterior derivative of M, then there is an isomorphism

$$\mathcal{H}^k(M) \cong H^k_{\text{dR}}(M)$$

where the left hand side is the space of Harmonic k-forms on M.

However, since M is (almost) complex, the exterior derivative splits into its holomorphic ∂ and its antiholomorphic $\overline{\partial}$ counterpart, and so does the cohomology groups. In this section we will show that Hodge theory for $\overline{\partial}$ also holds true.

6.1 Complex Properties of the Hodge Star Operator

Proposition 6.1.1

Let (M, h) be a complex Hermitian manifold. Let $n = \dim(M)$. Then the following are true.

- $*\omega \in \Omega^{(n-p,n-q)}$ for any $\omega \in \Omega^{p,q}(M)$.
- For any $\phi, \psi \in \Omega^{p,q}(M)$, we have

$$\langle \phi, \psi \rangle_{L^2} dV = \phi \wedge *\psi$$

- For any $\psi \in \Omega^{(p,q)}(M)$, we have $* * \phi = (-1)^{p+q}\psi$.

6.2 The Antiholomorphic Adjoint Differential Operator

Definition 6.2.1 (The Adjoint of $\overline{\partial}$)

Let (M, h) be a complex Hermitian manifold. Define the operator $\overline{\partial}^* : \Omega^{p,q}(M) \to \Omega^{p,q-1}(M)$ to be the operator such that

$$\langle \overline{\partial}^* \omega, \eta \rangle_{L^2} = \langle \omega, \overline{\partial} \eta \rangle_{L^2}$$

for all $\omega \in \Omega^{p,q}(M)$ and $\eta \in \Omega^{p,q-1}(M)$.

Proposition 6.2.2

Let (M, h) be a complex Hermitian manifold. Then the following are true.

- $\overline{\partial}^* = - * \overline{\partial} *$.
- $\overline{\partial}^* \circ \overline{\partial}^* = 0$.

Definition 6.2.3 (The Complex Laplacian Operator)

Let (M, h) be a complex Hermitian manifold. Define the complex Laplacian operator to be the map $\Delta_{\overline{\partial}} : \Omega^{p,q}(M) \to \Omega^{p,q}(M)$ by

$$\Delta_{\overline{\partial}} = \overline{\partial} \circ \overline{\partial}^* + \overline{\partial}^* \circ \overline{\partial}$$

Definition 6.2.4 (Complex Harmonic Forms)

Let M be a complex Hermitian manifold. Let $\omega \in \Omega^{p,q}(M)$ be a differential form. We say that ω is harmonic if $\omega \in \ker(\Delta_{\overline{\partial}})$.

Definition 6.2.5 (Space of Harmonic Forms)

Let M be a complex Hermitian manifold. Define the space of harmonic (p, q)-forms to be

$$\mathcal{H}^{p,q}(M) = \ker(\Delta_{\overline{\partial}})$$

Theorem 6.2.6 (Complex Hodge Theorem)

Let M be a complact complex Hermitian manifold. Then the following are true.

- $\mathcal{H}^{p,q}(M)$ is finite dimensional for all $p, q \in \mathbb{N}$.

- There is a decomposition

$$\Omega^{p,q}(M) \cong \mathcal{H}^{p,q}(M) \oplus \overline{\partial}(\Omega^{p,q-1}(M)) \oplus \overline{\partial}^*(\Omega^{p,q+1}(M))$$

Corollary 6.2.7
Let M be a compact complex Hermitian manifold. Then the quotient map induces an isomorphism

$$\mathcal{H}^{p,q}(M) \cong H^{p,q}(M)$$

Proposition 6.2.8
Let M be a compact complex Hermitian manifold. Let $n = \dim(M)$. Then the following are true.
- $H^{n,n}(M) = \mathbb{C}$.
- There is a perfect pairing $H^{p,q}(M) \times H^{n-p,n-q}(M) \to \mathbb{C}$ given by

$$(\alpha, \beta) \mapsto \int_M \alpha \wedge \beta$$

7 Kahler Manifolds

7.1 Properties of Kahler Manifolds

Definition 7.1.1 (Kahler Forms)
Let M be a complex Hermitian manifold. Let ω be the associated form. We say that ω is a Kahler form if $d\omega = 0$.

Definition 7.1.2 (Kahler Manifolds)
A Kahler manifold is a complex Hermitian manifold M together with a Kahler form ω.

Note: This makes Kahler manifolds also symplectic manifolds.

Proposition 7.1.3
Let (M, g) be a Riemannian manifold of dimension $2n$. Then M admits a Kahler form if and only if $\text{Hol}(\nabla^g) \leq U(n)$.

Example 7.1.4
$\mathbb{CP}^n$ together with the Fubini-Study metric is a Kahler manifold.

Example 7.1.5
Every Riemann surfaces is a Kahler manifold.

Theorem 7.1.6 (Hodge Decomposition Theorem)
Let M be a compact Kahler manifold. Then the following are true.

- There is an isomorphism
$$H^k_{\text{dR}}(M; \mathbb{C}) \cong \bigoplus_{p+q=k} H^{p,q}(M)$$
for all $k \in \mathbb{N}$.
- There is an isomorphism
$$H^{p,q}(M) \cong \overline{H^{q,p}(M)}$$
for all $p, q \in \mathbb{N}$.
- $\dim_{\mathbb{R}}(H^k_{\text{dR}}(M))$ is even for all $k \in \mathbb{N}$.

Example 7.1.7
Every compact Riemann surface is of the form Σ_g for some $g \geq 0$. We have seen above that Σ_g is Kahler. We know that
$$H^k_{\text{dR}}(\Sigma_g; \mathbb{C}) = \begin{cases} \mathbb{C} & \text{if } k = 0, 2 \\ \mathbb{C}^{2g} & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$
From this we conclude that $H^{0,0}(\Sigma_g) = \mathbb{C}$. Now since $\dim_{\mathbb{C}}(H^{p,q}(M)) = \dim_{\mathbb{C}}(\overline{H^{q,p}(\Sigma_g)}) = \dim_{\mathbb{C}}(H^{q,p}(\Sigma_g))$, we conclude that $H^{1,0}(\Sigma_g) \cong H^{0,1}(\Sigma_g) \cong \mathbb{C}^g$. Also, since Σ_g has complex dimension 1, it has no holomorphic 2-forms. Hence $H^{2,0}(\Sigma_g) = H^{0,2}(\Sigma_g) = 0$. This leaves us with $H^{1,1}(\Sigma_g) = H^2(\Sigma_g) = \mathbb{C}$. The hodge diamond of Σ_g is given by
$$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$$

Definition 7.1.8 (Hodge Filtration)
Let M be a compact Kahler manifold. Define the Hodge filtration of M to be

$$F^iH^k_{\text{dR}}(M;\mathbb{C}) = \bigoplus_{p\geq i} H^{p,q}(M)$$

Definition 7.1.9 (The Lefschetz Operator)
Let M be a Kahler manifold. Let ω be its associated form. Define the Lefschetz operator to be the map $L : \Omega^k(M) \to \Omega^{k+2}(M)$ given by

$$\alpha \mapsto \alpha \wedge \omega$$

Theorem 7.1.10 (Hard Lefschetz Theorem)
Let M be a compact Kahler manifold. Then the following are true.

- The Lefschetz operator induces an isomorphism

$$L^k : H^{n-k}_{\text{dR}}(M) \xrightarrow{\cong} H^{n+k}_{\text{dR}}(M)$$

Definition 7.1.11 (Primitive Cohomology)
Let M be a compact Kahler manifold. Let $n = \dim(M)$. Define the primitive cohomology of M to be

$$H^i_{\text{prim}}(M) = \ker(L^{n-i+1} : H^i_{\text{dR}}(M) \to H^{2n-i+2}_{\text{dR}}(M))$$

Definition 7.1.12 (Lefschetz Decomposition)
Let M be a compact Kahler manifold. Define the Lefschetz decomposition of M to be

$$H^k_{\text{dR}}(M) = \bigoplus_{2r\leq k} L^r(H^{k-2r}_{\text{prim}}(M))$$

Proposition 7.1.13
Let M be a compact Kahler manifold of complex dimension $2n$. Suppose that $h = \sum_{p=0}^{2m}\sum_{q=0}^{2m}(-1)^p \dim_{\mathbb{C}}(H^{p,q}(M))$. Then the signature of the non-degenerate symmetric bilinear form $H^n(X;\mathbb{R}) \times H^n(X;\mathbb{R}) \to \mathbb{R}$ from Poincare duality is given by $(h, 2n - h)$.

7.2 Line Bundles on Kahler Manifolds

Theorem 7.2.1 (Lefschetz Theorem on $(1,1)$-Classes)
Let M be a compact Kahler manifold. Then we have

$$\text{im}(\text{Pic}(M) \to H^2(M,\mathbb{Z}) \to H^2(M,\mathbb{C})) = \text{im}(H^2(M,\mathbb{Z}) \to H^2(M,\mathbb{C})) \cap H^{1,1}(M)$$